Week Six Assignment Louis Desgrée: Web Servers in Your Environment

In this lab report, I will demonstrate how to implement a vulnerable web server using Damn Vulnerable Web Application (DVWA) within VirtualBox. The instructions are based on the Chapter 4 readings and focus on setting up the web app on VMnet2 so that it remains isolated from other networks. By following these steps, you will create a functional DVWA installation and provide the required screenshot showing the vulnerable web server's name and IP address in the browser's address bar.

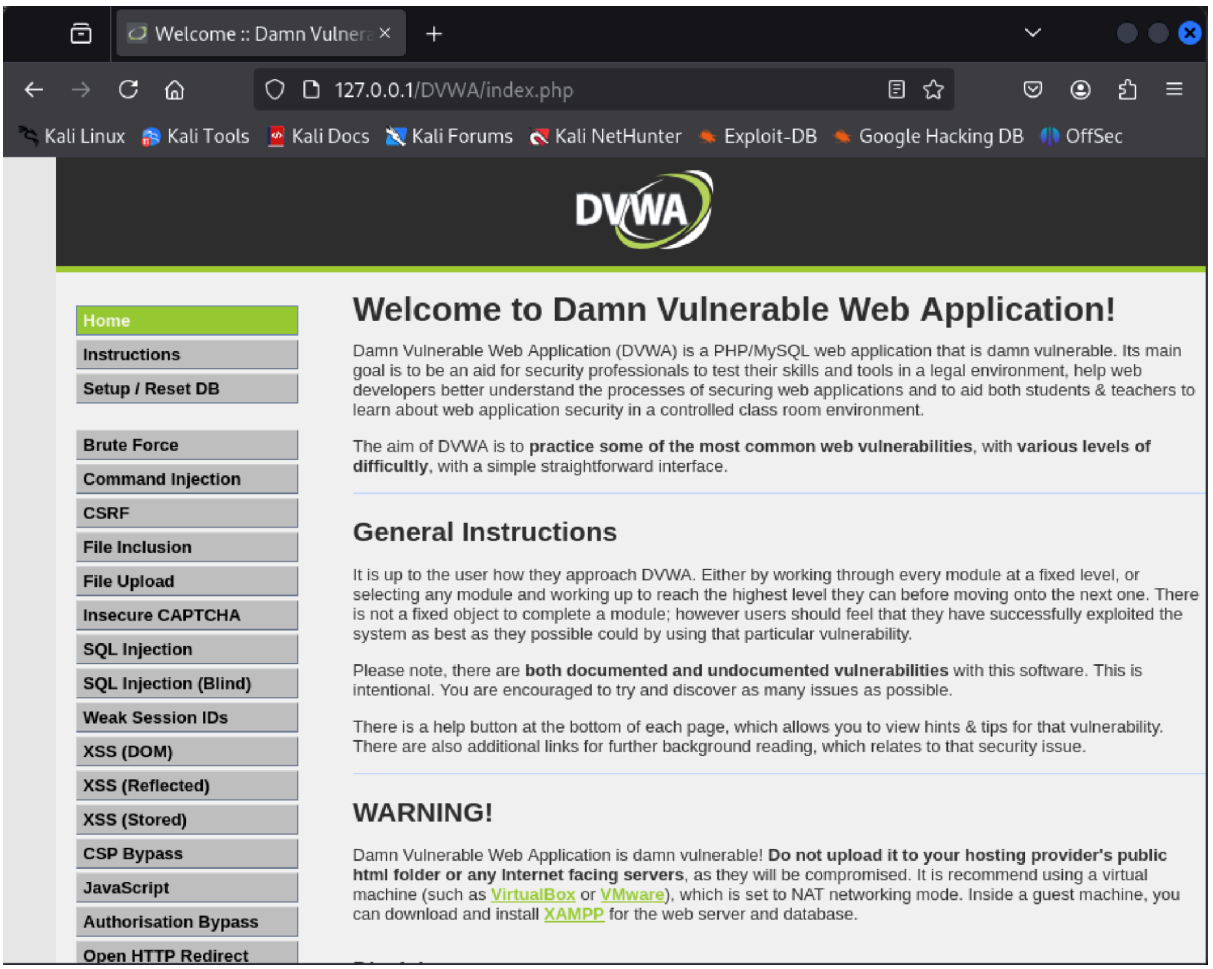
I began by preparing my environment in VirtualBox. First, I created a new virtual machine and configured its network adapter to use VMnet2, ensuring that the machine would be accessible on a private network. Next, I installed a Linux distribution (for example, Kali Linux) on this virtual machine. After installing the operating system, I updated all packages using the appropriate commands so that I would have the latest repositories and dependencies.

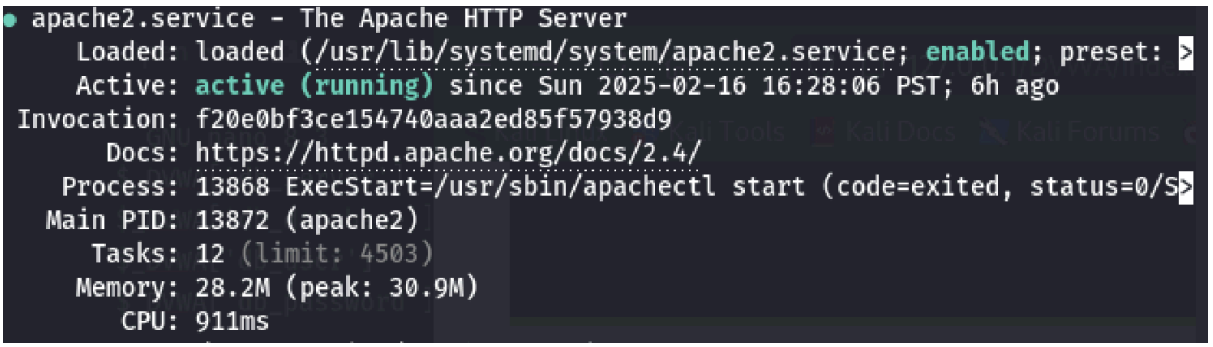
Once the system was updated, I installed Apache to serve web pages. I verified Apache's functionality by starting the service and confirming it was running without errors. With Apache running, I then proceeded to install MariaDB, which replaced the older MySQL server packages that were not available in my repositories. After installing MariaDB, I set a secure root password by logging into the database console and using the appropriate SQL commands. Exiting the database console was crucial before moving on to file-related tasks.

With the web server and database ready, I created a new database named "dvwa" by logging into MariaDB again and issuing the CREATE DATABASE dvwa; command. I then exited the database console to continue the installation process for DVWA itself. I navigated to the web root directory, which was located at /var/www/html on my Linux virtual machine, and cloned the DVWA repository from GitHub. This gave me a DVWA folder containing all necessary files. I copied the sample configuration file named config.inc.php.dist into a new file named config.inc.php and opened it with a text editor to update the database credentials. I set the db_server to 127.0.0.1 (or localhost), the db_user to root, and the db_password to the password I had configured in MariaDB.

To ensure that Apache could properly serve DVWA, I granted ownership of the DVWA folder to the Apache user by issuing the appropriate chown command. I restarted Apache to apply any changes and then opened my web browser on the host machine. In the address bar, I typed the IP address assigned to my virtual machine on VMnet2 (for instance, http://192.168.56.101/DVWA), which loaded the DVWA setup page. At this point, I clicked the "Create / Reset Database" button to initialize the DVWA tables and complete the setup.

After the database was created, DVWA prompted me to log in with the default credentials of "admin" for the username and "password" for the password. Upon successful login, I saw the main DVWA dashboard, confirming that the vulnerable web server was functioning correctly. Below is the screenshot requirement:





This screenshot clearly shows the IP address in the browser's address bar and the name of the vulnerable web application, meeting the assignment requirement. If you prefer to use Metasploitable or another vulnerable web application, the process is similar: configure the VM on VMnet2, install or import the vulnerable application, start the service, and then capture the screenshot of the main page with the IP address in the browser.

In conclusion, by carefully following these steps, I was able to set up DVWA on VirtualBox using VMnet2. This environment isolates the vulnerable web server from other networks, making it safer to experiment with common web vulnerabilities. I have documented each stage of the process, from configuring the network to verifying the final installation, and included the required screenshot. This lab report fulfills the assignment criteria of listing all steps, demonstrating a functioning vulnerable web server, and providing a clear visual of the final result.